

CatBoost Model

1. Model Overview

CatBoost (Categorical Boosting) is a gradient boosting model developed by Yandex. Its main advantage over other boosting models is that it handles categorical features natively — there is no need to manually encode them before training. It uses an internal technique called ordered target encoding, which is applied during training to avoid data leakage.

Like XGBoost, it builds trees sequentially where each tree corrects the errors of the previous one, but CatBoost uses symmetric (oblivious) trees which make it faster and less prone to overfitting.

2. Data Setup

CatBoost was trained on a separate version of the dataset that kept the original categorical columns as raw strings, instead of the encoded numeric format used for XGBoost. This allows CatBoost to handle them internally.

- Training set: catboost_data_train.csv
- Test/Validation split: catboost_data_test.csv was split 50/50 into validation and test sets using `random_state = 42`.
- Categorical features were detected automatically by selecting all object-type columns.

3. Hyperparameters Used

- iterations: 4000 — maximum number of trees to build.
- learning_rate: 0.005 — small value for slow, careful learning.
- depth: 8 — each tree can be up to 8 levels deep.
- loss_function: Logloss — suitable for binary classification.
- eval_metric: AUC — used to monitor performance during training.
- early_stopping_rounds: 50 — training stops automatically if the AUC on the validation set does not improve for 50 consecutive iterations.

4. Early Stopping

The model was set to train for up to 4000 iterations, but early stopping was used to prevent overfitting. Training stopped at iteration 3179 because the validation AUC did not improve for 50 rounds in a row. The model was then shrunk back to the best checkpoint at that iteration.

- Best validation AUC during training: 0.9256
- Best iteration: 3179 (out of 4000)
- Training accuracy at that point: 88.07%
- Validation accuracy at that point: 86.90%

5. Results on Test Set

- Accuracy: 87.48%
- AUC Score: 0.93
- Class 0 ($\leq 50K$) — Precision: 89%, Recall: 94%, F1: 92%
- Class 1 ($> 50K$) — Precision: 80%, Recall: 68%, F1: 74%

The model achieved a strong AUC of 0.93, which means it is very good at ranking and separating the two income classes.

Then it is the king of all used models